

Edufly – A Smart ERP Solution for Schools and Colleges

Presented by

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Introduction

Edufly is a full-stack academic management platform built with React, Node.js, and MySQL that streamlines institutional workflows through role-based dashboards for admins, teachers, and students. The system integrates secure authentication, student enrollment, course and class management, attendance tracking, result publishing, and real-time notifications. By combining use-case driven design with structured database models and workflow automation, Edufly reduces manual effort, improves transparency, and ensures efficient academic operations. The goal is to provide a unified digital environment where institutions can manage academics seamlessly.

Objective

- Automate institutional academic processes.
- Provide role-based dashboards for admins, teachers, and students.
- Ensure secure authentication and data management.
- Enhance efficiency in result publishing and course management.

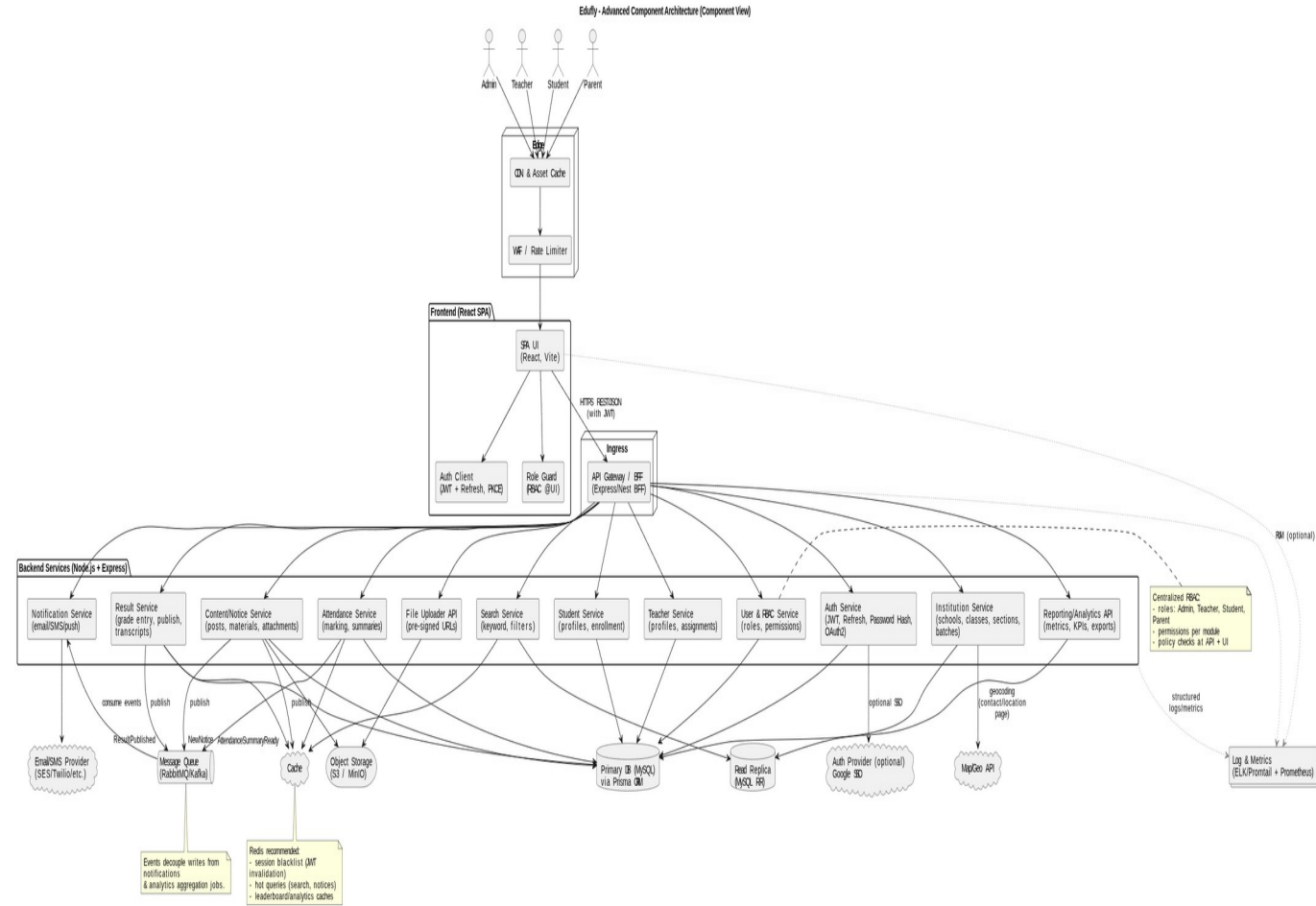
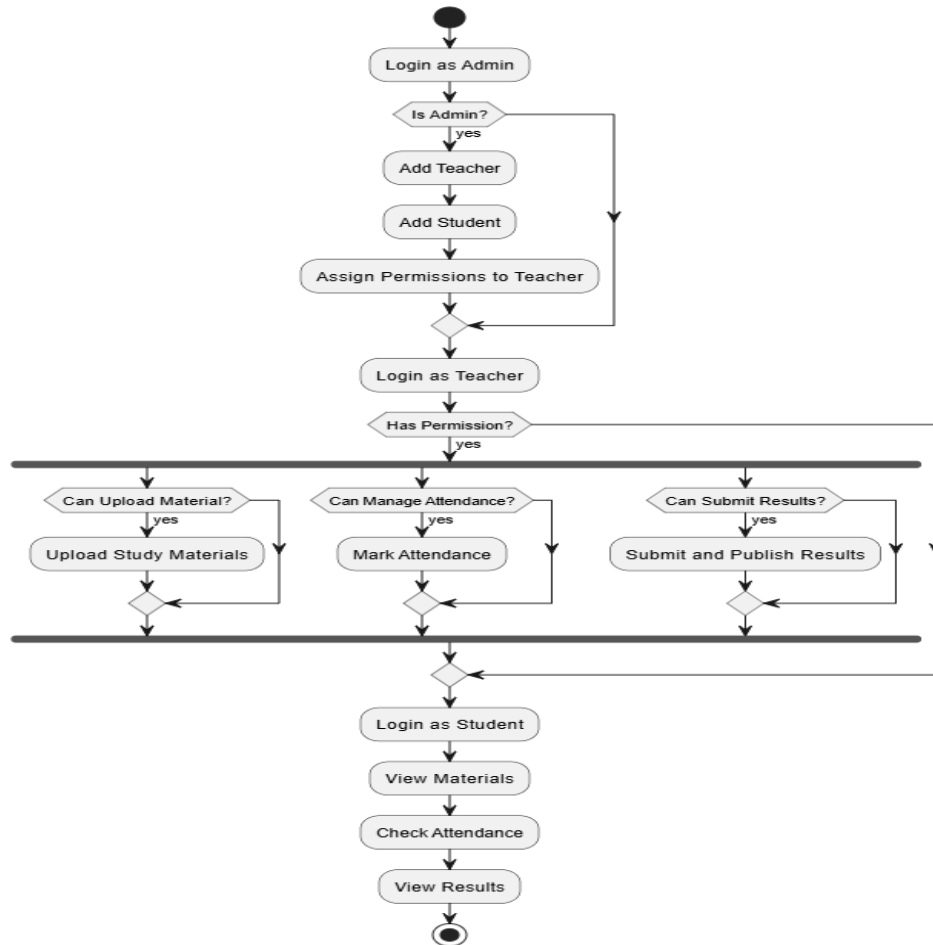
Background Study

Academic management in Bangladesh and beyond is still largely dependent on manual processes or fragmented digital tools, with many institutions using separate systems for attendance, results. Prior studies emphasize the importance of centralized platforms and role-based access control in ensuring efficiency, transparency, and reduced administrative overhead. Modern web stacks such as React (SPA), Node.js (REST APIs), and MySQL provide robust, scalable, and maintainable solutions at relatively low cost. A review of existing platforms shows that while some offer basic attendance or result modules, they lack integrated workflows that connect students, teachers, and administrators under one system. These insights shaped Edufly's architecture and role-based workflows, aiming to close the gap between isolated academic functions and a unified institutional management system.

Gap Analysis

Features	<u>Eduman</u>	<u>Fedena</u>	<u>MyClassCampus</u>	<u>OpenEduCat</u>	<u>Edufly</u>
Manage Students	Yes	Yes	Yes	Yes	Yes
Manage Teachers	Yes	Yes	Yes	Yes	Yes
Provide Website	No	No	No	No	Yes
Notice Board	No	No	Yes	Yes	Yes
Student Information System	Yes	Yes	Yes	Yes	Yes
Attendance Management	Yes	Yes	Yes	Yes	Yes
User and Role Management	No	No	No	No	Yes
Multi-role Authentication	No	No	No	No	Yes

Methodology



Result and Analysis

- Successfully implemented multi-role system: Admin, Teacher, Student.
- Automated workflows for result publishing, and notifications.
- Database model ensures data consistency.
- Sequence diagram shows role-based interactions (admin, teacher, student)

Nobility and Uniqueness

- Tailored for Bangladeshi schools/colleges
- Combines website + ERP in one platform
- Uses open-source stack → affordable, scalable
- Reduces paper use, supports sustainability

Conclusion

This project delivers a secure, modular academic management platform that transforms traditional manual processes into an automated, role-based digital workflow. It demonstrates reliable authentication, streamlined student enrollment, efficient course and result management, and effective communication through notifications, creating a unified academic ecosystem for institutions. By addressing the fragmentation in existing systems, Edufly improves transparency, reduces administrative workload, and enhances the overall learning experience. While multi-institution scalability remain as next steps, the present system establishes a strong, extensible foundation for real-world academic deployment and future enhancements.

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THANK YOU